

Midterm Exam

Version A

Friday, October 9, 2026 · in class

Name: _____ Student ID: _____

40 points | Suggested working time 40 minutes

Use a basic calculator. Show your working. Allow about 7, 10, 7, 10 and 3 minutes for Sections 1–5, plus 3 minutes to read and check. Sections 1 and 5 are marked on clear reasoning, not on matching one “right” answer. The data are synthetic teaching data from a fictional seaside café.

Section	What it covers	Points	Score
1	Choose the inputs	8	
2	Read the LINEST output	12	
3	Errors and outliers	8	
4	Did the price change sales?	9	
5	Summary for the owner	3	
Total		40	

SECTION 1 · Choose the inputs

8 points · about 7 minutes

A seaside café wants to predict **tomorrow's gelato sales** (cups) **before it opens**. Table 1 lists information it could use. Across 168 days of data, **temperature and sunshine are strongly related to each other ($r = 0.78$)**. Their correlations with gelato sales are 0.58 (temperature) and 0.49 (sunshine).

Table 1. Candidate information.

#	Information
1	Tomorrow's temperature forecast (°C)
2	Tomorrow's sunshine forecast (hours)
3	Whether tomorrow is a weekend (yes or no)
4	Whether a beach event is on the calendar tomorrow (yes or no)
5	Tomorrow's planned gelato price (€)
6	Tomorrow's iced coffee sales (cups), counted at closing time

1a Choose **three** inputs for the model. For each, say whether it should push gelato sales **up or down**, and give a one-line reason. 3 pts

Input (name or number)	Up or down?	Reason (one line)

1b One candidate in Table 1 is **unsuitable** for this forecast. Which one, and what is the problem? 2 pts

1c Should temperature **and** sunshine both go into the same model? Explain, and say which one you would keep. 2 pts

1d How would you check the model's prediction errors on days that were **not** used to fit it? 1 pt

SECTION 2 · Read the LINEST output

12 points · about 10 minutes

A model of daily gelato sales (cups) was fitted to 168 days with $\text{=LINEST}(y, X, \text{TRUE}, \text{TRUE})$. The three X columns were entered in this order: X_1 = **temperature (°C)**, X_2 = **weekend (1 = Saturday or Sunday, 0 = weekday)**, X_3 = **beach event (1 = event, 0 = no event)**. Use Table 2 to read Table 3.

Table 2. How LINEST lays out its results.

Col A (X_3)	Col B (X_2)	Col C (X_1)	Col D (Intercept)	Statistic description
Slope m_3	Slope m_2	Slope m_1	Intercept (b)	Coefficients for: $y = m_1X_1 + m_2X_2 + m_3X_3 + b$
SE – m_3	SE – m_2	SE – m_1	SE – Intercept	Standard error for each coefficient
R^2	SE – Y	Blank	Blank	Quality of the model fit
F-statistic	df	Blank	Blank	Analysis of variance metrics
SS – Reg	SS – Resid	Blank	Blank	Sum of squares metrics

Table 3. LINEST output for the model (shaded cells are blank).

Col A	Col B	Col C	Col D
25.74	34.16	5.57	48.97
4.27	3.85	0.52	11.86
0.61	22.52		
86.39	164		
131402.67	83146.74		

2a Write the fitted equation, using the numbers in Table 3 and the variable names.

2 pts

2b Each statement below misreads the output. Rewrite each one correctly, keeping the units and conditions.

3 pts

(i)	"The intercept means the café sells 48.97 cups on an average day."
(ii)	"Beach events cause the café to sell 25.74 more cups."
(iii)	" $R^2 = 0.61$ means the model's predictions are correct 61% of the time."

2c (a) By how much does predicted sales change on a day that is **5 °C warmer**, with the weekend and event values unchanged? (b) Which changes predicted sales more: a 5 °C rise or a beach event? (c) Why can't you compare the temperature coefficient with the event coefficient directly?

3 pts

2d Tomorrow is forecast to be 26 °C, on a weekend, with a beach event. (a) Predict gelato sales. (b) Actual sales turn out to be **215 cups**: calculate the residual. (c) SE – Y is the typical size of a forecast error. Is your residual smaller or larger than SE – Y, and what does that say about this forecast?

4 pts

SECTION 3 · Errors and outliers

8 points · about 7 minutes

3a · Errors. Sales and errors are in cups. These are constructed exercise numbers, not workbook output. **Residual = actual – predicted.** **MAE = (sum of the absolute residuals) ÷ n.**

Day	Actual	Predicted	Residual
Mon	180	190	
Sat		230	+15
Wed	160		–5

3a-i Fill the blanks in the table.

1 pt

3a-ii Calculate the MAE of these predictions.

1 pt

3a-iii Model Z has residuals of –6, –6, –6 on the same three days. Which model has the lower MAE? What do the identical signs tell you about Model Z?

2 pts

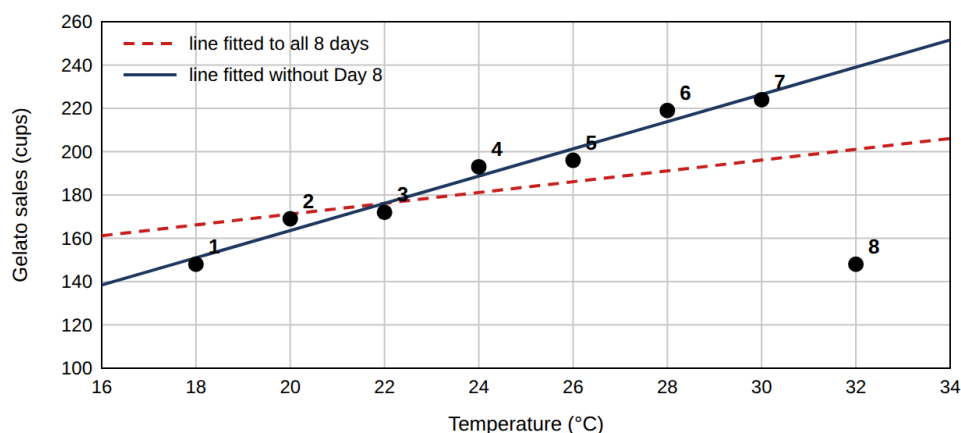
3b · Outliers. Table 4 and Figure 1 show gelato sales on 8 days (constructed example). Table 5 gives the straight-line results with and without Day 8.

Day	1	2	3	4	5	6	7	8
Temperature (°C)	18	20	22	24	26	28	30	32
Sales (cups)	148	169	172	193	196	219	224	148

Table 5. Results for the straight line (sales against temperature).

	All 8 days	Without Day 8
Slope (cups per °C)	2.49	6.29
R ²	0.17	0.97
SE – Y (cups)	28.77	5.18

Figure 1. The 8 days, with the line fitted to all of them (dashed) and the line fitted without Day 8 (solid).



3b-i Which day does not follow the pattern of the others?

1 pt

3b-ii Compare the results with and without that day. What impact does one day have on the slope and on R²?

2 pts

3b-iii Before deleting it, what would you do?

1 pt

SECTION 4 · Did the price change sales?

9 points · about 10 minutes

On 30 June the café raised the gelato price from **€4.00** to **€4.50**. Table 6 shows gelato sales (cups) on 5 weekdays **before** (Group 1) and 5 weekdays **after** (Group 2). Both samples are weekdays with similar weather (about 24 °C).

Formulas (as given in class):

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

A negative t only means that Group 2 has the larger mean.

Table 6. Worksheet (fill in the blank cells).

Day	Before (x ₁)	x ₁ - $\bar{x}_1$	(x ₁ - $\bar{x}_1$) ²	After (x ₂)	x ₂ - $\bar{x}_2$	(x ₂ - $\bar{x}_2$) ²
1	196			184		
2	204			192		
3	188			176		
4	212			196		
5	200			182		
Sum						
Mean $\bar{x}$						
s						

4a Complete Table 6: the two means, the two sums of squared deviations, and s for each group (2 decimals). **3 pts**

4b Calculate s_p and t (2 decimals). **2 pts**

Sheets gives **=T.TEST(B2:B6, C2:C6, 2, 2) = 0.031** (two-tailed, equal variances).

4c (i) State the null hypothesis in words. (ii) Is the difference statistically significant at 5%? Explain using the p-value. (iii) In plain words, what does p = 0.031 mean? **3 pts**

4d Name one other change, besides the price, that could explain the drop in sales. **1 pt**

SECTION 5 · Summary for the owner

3 points · about 3 minutes

5 Write 3–4 sentences to the café owner, who has no statistics background. Include: (i) which inputs you recommend for forecasting gelato sales and the main finding from the LINEST output; (ii) what the price test showed; (iii) one caution. **3 pts**

— End of exam —